IREDDI RAKSHITHA

Software Development Engineer | ireddirakshitha@gmail.com | +91 8008131387 | LinkedIn | GitHub

EDUCATION

Vellore Institute of Technology *Aug 2021 – May 2025*
B.Tech in Computer Science and Engineering; CGPA: 9.4/10

WORK EXPERIENCE

Software Development Engineer I - Amazon, Bangalore *Sep 2026 – Present*

- Developing **Java/Spring** backend services for Amazon's Identity Verification (KYC/KYB) platform, supporting **4.8M+ customers and businesses** across payments, lending, seller onboarding, and hiring.
- Contributing to KYC globalization across **APAC, EU, and NA**, building configuration-driven, region-aware microservices using **AWS, DynamoDB, S3, KMS, CloudHSM, Lambda, Step Functions, SQS, and Rekognition** for document validation, e-KYC, and biometric verification with field-level encryption and secure data handling.

Software Engineer - Barclays, Pune *Jul 2025 – Sep 2026*

- Led cloud modernization and legacy SAS-to-AWS migration initiatives within RDS, driving system integration and collaboration across engineering and business teams.
- Delivered migration of **15+ customer-facing credit decisioning services** from Triad to PCSM, developing hybrid integration frameworks and AI-assisted workflow analysis and validation for end-to-end production releases.
- Led **Java 8-to-21 modernization** and built migration automation using **Python, Spark, Databricks, and GitLab**, reducing processing time by **45%** across concurrent migration workstreams.

SDE Intern - WorkIndia, Bangalore *Jan 2025 – Jun 2025*

- Developed Python-based backend services and workflow automation tools supporting internal business-critical processes. Designed and implemented Kafka-based data pipelines for processing high-volume application events and enabling reliable downstream analytics workflows.
- Partnered with product and business stakeholders to build reporting and analytics solutions that supported growth tracking, operational insights, and strategic decision-making.

SDE Intern - Barclays, Pune *May 2024 – Jul 2024*

- Supported DevOps and automation activities for Risk, Finance, and Treasury platforms, ensuring reliable execution of enterprise workflows and scheduled processes.
- Built and maintained Python and Kedro-based data processing workflows that generated financial reports consumed by downstream reporting and analytics systems.
- Contributed to the Hyperion GFT project through workflow integration, production support, testing, validation, and deployment activities for critical enterprise finance applications.

SKILLS SUMMARY

- **Programming & Backend:** Java, Python, C++, C, SQL, DSA, System Design, Spring Boot, FastAPI, REST APIs, Multithreading, Git
- **Distributed Systems & Cloud:** Distributed Systems, Microservices, Kafka, AWS, Azure, DynamoDB, S3, KMS, Lambda, Step Functions, Docker, Kubernetes, CI/CD
- **Data Engineering:** Apache Spark, PySpark, Databricks, ETL Pipelines, Hadoop, Hive, PostgreSQL, MongoDB, Redis
- **AI & Machine Learning:** LLMs, LLM Agents, Agentic AI, RAG, Transformers, PyTorch, Hugging Face, LangChain, MCP
- **Software Engineering:** Design Patterns, API Design, Unit Testing, Integration Testing, Workflow Automation

PROJECTS

- **FinTrack – Finance and Expense Intelligence Platform:** Built a full-stack financial management platform with expense tracking, budgeting, and automated spending insights, including authentication, transaction APIs, and real-time dashboards.
  Tech: Java, Spring Boot, React, PostgreSQL, JWT, REST APIs, Docker, AWS
- **SecureAuth - Identity and Access Management Service:** Built a centralized IAM platform with role-based access control, multi-factor authentication, and token-based session security, backed by scalable, cached APIs for secure multi-application access.
  Tech: Spring Boot, Spring Security, Redis, PostgreSQL, Docker, Kubernetes, REST APIs
- **DataPulse - Real-Time Customer Analytics and Recommendation Engine:** Built a real-time customer analytics platform using Kafka ingestion pipelines and Spark processing to generate trend analysis and personalized recommendations at scale.
  Tech: Python, Kafka, PySpark, Airflow, PostgreSQL, MongoDB, Docker, ETL
- **InsightAI - Enterprise Knowledge Assistant :** Built an AI-powered RAG assistant that answers questions from enterprise documents using HuggingFace embeddings and FAISS/ChromaDB vector search, with conversational memory for context-aware responses.
  Tech: Python, LangChain, OpenAI API, HuggingFace Transformers, FAISS, ChromaDB, FastAPI, PostgreSQL

PUBLICATIONS

TRACE: Verifiable RL Environment for Network Troubleshooting Agents — *SLM-Agents: NeurIPS Workshop 2026*
Contributed to the development and evaluation of a verifiable reinforcement learning environment for autonomous network troubleshooting agents.

APISHIFT: Verifiable Agent Learning for API Contract Migration — *AI for Verifiable Coding: NeurIPS Workshop 2026*
Contributed to benchmark design and evaluation workflows for agent-based API migration and software engineering tasks.

MVEB: Massive Video Embedding Benchmark — arXiv (2026)
Introduced MVEB, a 23-task benchmark for video embeddings spanning classification, clustering, retrieval, and video-centric question answering; evaluated 33 models and released the benchmark, code, and leaderboard, integrating into the MTEB ecosystem for unified text/image/audio/video evaluation.

Additional Publications:

- *Resolving Spatial and Semantic Conflicts in Multi-LoRA Composition via Softmax Routing and Cross-Adapter Attention Conditioning* — **ICLR'27**
- *Through the Looking Glass: Benchmarking Mirror Image Reasoning in Vision-Language Models* - **CVPR'26**
- *Reproducible, Cross-Platform Study of Order-Book Data Structures in a Low-Latency Matching Engine*
- *CRYO-AGENT: Agentic Cryo-ET Reconstruction with Physics-Guided Retrieval-Augmented Diffusion* - Ongoing
- *TabuLM: Morphology-Aware Tabular Pre-training for Low-Resource Languages*
- *ML-Based SDN DDoS Detection: Machine Learning–Enabled Intrusion Detection System* - **CIMA'26**
- *Physics-Informed Neural Network Analysis of Rotational Effects on Oscillatory MHD Flow* - Journal Scopus
- *Comparative Analysis of Machine Learning Algorithms for Cervical Cancer Prediction* - **Springer Nature · ICACECS 2023**
- *Revolutionizing Solar Generation Data Mining through Advanced Machine Learning Algorithms* - **CSITSS 2023**
- *Anime Visage: GAN-Assisted Character Development* - **ICOECA 2024**
- *KinyaEmbed: Contrastive Sentence Embeddings for Kinyarwanda via Multi-Stage Curriculum Training*

Research Contributions: Benchmarking, dataset construction, and evaluation frameworks for vision-language models, LLM agents, agentic AI systems, software engineering agents, and verifiable reinforcement learning environments.

PATENT

Filed an Indian patent titled **"A System to Have Adaptive Cruise Control for Vehicles"** under VIT-AP University, focusing on AI-driven intelligent vehicle speed control and automated decision-making based on real-time driving and traffic conditions. The patent was filed on 25 July 2024 and published under Section 11A, contributing to research in intelligent transportation systems and applied AI for smart automotive technologies.

ACHIEVEMENTS & HACKATHONS

- **Gold Medalist** - Ranked among the Top 10 students across all B.Tech programs at VIT-AP University and awarded the Gold Medal for academic excellence in Computer Science and Engineering - Data Science & Analytics.
- **Barclays Recognition** - Received Director's Recognition for rapid learning, domain expertise, and early contributions within the Risk Decisioning Services team, including support for the successful Triad to PCSM migration.
- **Hackathon Achievements** - Top 100 Finalist, *Meta Open Source AI Hackathon*; Winner, *Elicit'23: UX Evolve* (Manipal University Jaipur); Finalist, *Great India HackFest 2024* (CSI Chapter, VIT-AP University).
- **Innovation Award** - Developed an ECG Monitoring System using Raspberry Pi, recognized as the winning project among 500+ teams for technical innovation and implementation excellence. Awarded project funding for further development.

VOLUNTEER EXPERIENCE

Technical Committee Member - IEM-HEALS 2026 *2026*
Selected as a Technical Committee Member for the International Conference on Advancing Science and Technologies in Health Science (IEM-HEALS 2026), supporting the peer-review process and technical evaluation of research papers.

Community Manager - CSI VIT-AP Chapter *Nov 2021 – Aug 2024*
Progressed from Coordinator to Lead to Community Manager, leading technical events, hackathons, AI workshops, and project-based learning sessions; mentored juniors in programming, AI/ML projects, research, and development activities. Awarded Best Community Manager and Best Team Award for leadership and community impact, supporting and mentoring 500+ students through technical events, AI initiatives, and collaborative learning programs.